SINGLE IMAGE DEHAZING

PROJECT SUMMARY

This project will compare CLAHE, coarse Dark Channel Prior (DCP), and DCP with guided filtering to restore visibility in hazy images. We will use 30 RESIDE scene pairs: 10 for parameter selection and 20 for final testing. Evaluation will compare PSNR, SSIM, runtime and visual artifacts. The four images below demonstrate the proposed processing stages on one sample.

Requirements & Libraries

Python / NumPy: Pixel arrays and numerical operations.
OpenCV: Image handling, local filtering and CLAHE.
scikit-image: PSNR and SSIM evaluation.
Matplotlib: Comparison figures and result tables.



01 Original hazy image

Load an RGB image and normalize values to [0,1]. Haze reduces contrast and introduces an atmospheric light veil.



02 Dark channel

Take the minimum across RGB channels and a local window. Dark-channel candidates help estimate atmospheric light A.



03 Refined transmission

Refine the transmission estimate using image-guided filtering. Brighter pixels indicate higher transmission.



04 Restored image

Recover scene radiance using A and refined t. A lower bound limits noise amplification. Single-image demonstration.

MATHEMATICAL STEPS

1. DARK CHANNEL

$$D(I)(x) = \min I_c(y)$$

$y \in \Omega(x)$, $c \in \{R,G,B\}$

2. TRANSMISSION ESTIMATE

$$\tilde{t}(x) = 1 - \omega\, D(I/A)(x)$$

$\omega = 0.95$

3. GUIDED FILTER

$$q_i = a_k g_i + b_k$$

g: image guidance q: refined t

4. RADIANCE RECOVERY

$$J = (I - A) / \max(t, t_0) + A$$

$t_0 = 0.10$

Demo: authors' forest input, people.csail.mit.edu/kaiming/cvpr09/; computed here using a 15×15 patch, guided radius 20 and ε = 0.001. No RESIDE benchmark results yet.
Sources: He et al., CVPR 2009 (DCP); He et al., ECCV 2010 (guided filter); Li et al., IEEE TIP 2019 (RESIDE). Scope: classical image restoration, no model training.